

Numerical Methods for Non-Markovian Quantum State Diffusion

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Jan. 30, 2026

Group Meeting

Non-Markovian Quantum State Diffusion

Non-Markovian Quantum State Diffusion (NMQSD/NMSSE):¹

$$\frac{d}{dt}|\psi_t(z^*)\rangle = \left(-iH + Lz_t^* - L^\dagger \int_0^t \alpha(t,s) \frac{\delta}{\delta z_s^*} ds \right) |\psi_t(z^*)\rangle, \quad (1)$$

where

- H is the system Hamiltonian.
- L is the system-environment coupling operator.
- z_t^* is a complex noise, satisfying $\mathbb{E}[z_t] = 0$, $\mathbb{E}[z_t z_s] = 0$ and $\mathbb{E}[z_t z_s^*] = \alpha(t, s)$.
- $\alpha(t, s)$ is the bath correlation function.
- Reduced density matrix: $\rho_t = \mathbb{E}[|\psi_t(z^*)\rangle\langle\psi_t(z^*)|]$.

The state $|\psi_t(z^*)\rangle$ depends on the entire noise history $\{z_s^* | s \in [0, t]\}$!

¹L. Di'osi and W. T. Strunz, Phys. Lett. A, 235: 569–573 (1998).

Non-Markovian Quantum State Diffusion

Variational Derivative Term:

$$\int_0^t \alpha(t, s) \frac{\delta \psi_t(z^*)}{\delta z_s^*} ds := \left. \frac{d}{d\epsilon} \psi_t(z^* + \epsilon \alpha(t, \cdot)) \right|_{\epsilon=0}, \quad (2)$$

where $z^* = \{z_s^* | s \in [0, T]\}$ and $\alpha(t, \cdot) = \{\alpha(t, s) | s \in [0, T]\}$.

Discrete Form: Let $t_j = j\Delta t$, $j = 0, 1, \dots, N$, with $\Delta t = T/N$, then

$$\int_0^t \alpha(t, s) \frac{\delta \psi_t(z^*)}{\delta z_s^*} ds \approx \vec{\alpha} \cdot \nabla_{z^*} \psi_t(z^*), \quad (3)$$

where $\vec{\alpha} = [\alpha(t, t_0), \alpha(t, t_1), \dots, \alpha(t, t_N)]$ and $\nabla_{z^*} = \left[\frac{\delta}{\delta z_{t_0}^*}, \frac{\delta}{\delta z_{t_1}^*}, \dots, \frac{\delta}{\delta z_{t_N}^*} \right]$.

Hierarchy of Pure States (HOPS)

Consider a bath correlation function that can be expressed as an exponential decomposition (**Necessary**):

$$\alpha(t, s) = g e^{-\omega|t-s|}, \quad \text{or} \quad \alpha(t, s) = \sum_{j=1}^J g_j e^{-\omega_j|t-s|}. \quad (4)$$

Auxiliary Wave Functions

- Introduce a hierarchy of auxiliary wave functions $\psi_t^{(n)}$ via the operator D_t :

$$\psi_t^{(n)} = D_t \psi_t^{(n-1)}, \quad D_t = \int_{-\infty}^{\infty} \alpha(t, s) \frac{\delta}{\delta z_s^*} ds \quad (5)$$

- The time derivative of D_t satisfies:

$$\partial_t D_t = -\omega D_t \quad (6)$$

Hierarchy of Pure States (HOPS)

By applying the auxiliary operators, the linear NMQSD equation is transformed into a hierarchy of coupled equations.

$$\begin{aligned}
 \partial_t \psi_t^{(1)} &= \partial_t (D_t \psi_t^{(0)}) \\
 &= \partial_t D_t \psi_t^{(0)} + D_t \partial_t \psi_t^{(0)} \\
 &= -\omega D_t \psi_t^{(0)} + D_t (-iH + Lz_t^* - L^\dagger D_t) \psi_t^{(0)} \\
 &= -\omega \psi_t^{(1)} - iH \psi_t^{(1)} + LD_t z_t^* \psi_t^{(0)} - L^\dagger \psi_t^{(2)} \\
 &= (-iH - \omega + Lz_t^*) \psi_t^{(1)} + \alpha(0)L\psi_t^{(0)} - L^\dagger \psi_t^{(2)},
 \end{aligned} \tag{7}$$

where

$$\begin{aligned}
 D_t(Lz_t^* \psi_t^{(0)}) &= LD_t z_t^* \psi_t^{(0)} + Lz_t^* D_t \psi_t^{(0)} \\
 &= \alpha(0)L\psi_t^{(0)} + Lz_t^* \psi_t^{(1)}.
 \end{aligned} \tag{8}$$

Hierarchy of Pure States (HOPS)

By repeating this process, the coupled stochastic equations for an infinite hierarchy of pure states (HOPS)² $\{\psi_t^{(k)}\}$ are obtained:

$$\partial_t \psi_t^{(k)} = (-iH - k\omega + Lz_t^*) \psi_t^{(k)} + k\alpha(0)L\psi_t^{(k-1)} - L^\dagger \psi_t^{(k+1)}. \quad (9)$$

with the initial conditions:

$$\psi_0^{(0)} = \psi_0, \quad \psi_0^{(k)} = 0, \quad k \geq 1. \quad (10)$$

Terminator:

$$\psi_t^{(K+1)} \approx \frac{\alpha(0)}{\omega} L\psi_t^{(K)}, \quad \text{or} \quad \psi_t^{(K+1)} \approx 0. \quad (11)$$

²D. Suess, A. Eisfeld and W. T. Strunz, Phys. Rev. Lett., 113: 150403 (2014).

Interaction Picture and Reduced Form

Consider the interaction picture with respect to the bath:

$$\tilde{\psi}_t(z^*) = e^{iHt}\psi_t(z^*), \quad (12)$$

where H is the bath Hamiltonian. The NMQSD equation in the interaction picture reads:

$$\frac{d}{dt}\tilde{\psi}_t = \left(L(t)z_t^* - L^\dagger(t) \int_0^t \alpha(t,s) \frac{\delta}{\delta z_s^*} ds \right) \tilde{\psi}_t, \quad (13)$$

where $L(t) = e^{iHt}Le^{-iHt}$.

Analytical Solution

Theorem

The formal solution of the reduced NMQSD equation (13) can be expressed as:

$$\psi_t = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \psi_t^{(n,m)} = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} (-1)^m \int_{0 < \tau_1 < \dots < \tau_n < t} d^n \tau \int_{0 < s_1 < \dots < s_{2m} < t} f(\tau, \mathbf{s}) d^{2m} \mathbf{s},$$

$$f(\tau, \mathbf{s}) = \left(\prod_{k=1}^n z_{\tau_k}^* \right) \mathcal{L}_b(s_1, \dots, s_{2m}) \times \mathcal{T} \left(\prod_{k=1}^n L(\tau_k) \prod_{j=1}^m \left[L^\dagger(s_{P(2j)}) L(s_{P(2j-1)}) \right] \right) \psi_0. \quad (14)$$

Here

- $\mathcal{L}_b(s_1, \dots, s_{2m}) = \sum_P \prod_{i=1}^m \alpha(s_{P(2i)}, s_{P(2i-1)})$ is the bath correlation function contribution from all pairings P of the $2m$ time points, The pairing P in $\mathcal{L}_b$ connects a later time s_{later} with an earlier time $s_{earlier}$ such that $\alpha(s_{later}, s_{earlier})$ is formed.
- $\mathcal{T}$ denotes time-ordering of the operators.
- $d^n \tau = d\tau_1 \dots d\tau_n$ and $d^{2m} \mathbf{s} = ds_1 \dots ds_{2m}$.

Consistency

Lemma

The solution (14) satisfies the following equation:

$$\partial_t \psi_t^{(n,m)} = z_t^* L(t) \psi_t^{(n-1,m)} - L^\dagger(t) \int_0^t \alpha(t,s) \frac{\delta \psi_t^{(n+1,m-1)}}{\delta z_s^*} ds, \quad (15)$$

Proof:

Consider an equivalent form with the integration over the cube $C_n = [0, t]^n$:

$$\begin{aligned} \psi_t(z^*) &= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \psi_t^{(n,m)} \\ &= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} (-1)^m \int_{C_n} \frac{1}{n!} d^n \tau \int_{0 < s_1 < \dots < s_{2m} < t} f(\tau, s) d^{2m} s. \end{aligned} \quad (16)$$

Denote

$$\partial_t \psi_t^{(n,m)} = A_{n,m} + B_{n,m}, \quad (17)$$

where $A_{n,m}$ corresponds to the boundary term with $\tau_k = t$ for any $k = 1, \dots, n$, and $B_{n,m}$ corresponds to the boundary term with $s_{2m} = t$.

Consistency

For $A_{n,m}$, by reranging the remaining terms, we have

$$\begin{aligned}
 A_{n,m} &= (-1)^m \int_{C_{n-1}} \frac{1}{(n-1)!} d^{n-1} \tau \int_{0 < s_1 < \dots < s_{2m} < t} f_A(\tau, \mathbf{s}) d^{2m} \mathbf{s}, \\
 f_A(\tau, \mathbf{s}) &= \left(\prod_{k=1}^{n-1} z_{\tau_k}^* \cdot z_t^* \right) \mathcal{L}_b(s_1, \dots, s_{2m}) \\
 &\quad \times \mathcal{T} \left(\prod_{k=1}^{n-1} L(\tau_k) \cdot L(t) \prod_{j=1}^m [L(s_{P(2j-1)}) L^\dagger(s_{P(2j)})] \right) \psi_0.
 \end{aligned} \tag{18}$$

It derives

$$A_{n,m} = z_t^* L(t) \psi_t^{(n-1,m)}. \tag{19}$$

Consistency

For $B_{n,m}$, we have

$$\begin{aligned}
 B_{n,m} &= (-1)^m \int_{C_n} \frac{1}{n!} d^n \boldsymbol{\tau} \int_{0 < s_1 < \dots < s_{2m-1} < t} f_B(\boldsymbol{\tau}, \mathbf{s}) d^{2m-1} \mathbf{s}, \\
 f_B(\boldsymbol{\tau}, \mathbf{s}) &= \left(\prod_{k=1}^n z_{\tau_k}^* \right) \mathcal{L}_b(s_1, \dots, s_{2m-1}, t) \\
 &\quad \times \mathcal{T} \left(\prod_{k=1}^n L(\tau_k) \prod_{j=1}^{m-1} [L(s_{P(2j-1)}) L^\dagger(s_{P(2j)})] L(s_{P(2m-1)}) L^\dagger(t) \right) \psi_0.
 \end{aligned} \tag{20}$$

In pairing P in $\mathcal{L}_b(s_1, \dots, s_{2m-1}, t)$, the time point t must be paired with one of the earlier time points s_l ($l = 1, \dots, 2m-1$). Thus, we have

$$\mathcal{L}_b(s_1, \dots, s_{2m-1}, t) = \sum_{l=1}^{2m-1} \alpha(t, s_l) \mathcal{L}_b(s_1, \dots, s_{l-1}, s_{l+1}, \dots, s_{2m-1}). \tag{21}$$

Consistency

Consider the functional derivative term in the RHS:

$$\begin{aligned} \frac{\delta \psi_t^{(n+1, m-1)}}{\delta z_p^*} &= (-1)^{m-1} \int_{C_{n+1}} \frac{1}{(n+1)!} d^{n+1} \tau \int_{0 < s_1 < \dots < s_{2m-2} < t} f_\delta(\tau, s, p) d^{2m-2} s, \\ f_\delta(\tau, s, p) &= \sum_{l=1}^{n+1} \delta(\tau_l - p) \left(\prod_{k \neq l} z_{\tau_k}^* \right) \mathcal{L}_b(s_1, \dots, s_{2m-2}) \\ &\quad \times \mathcal{T} \left(\prod_{k=1}^{n+1} L(\tau_k) \prod_{j=1}^{m-1} \left[L(s_{P(2j-1)}) L^\dagger(s_{P(2j)}) \right] \right) \psi_0, \end{aligned} \quad (22)$$

Plugging into the memory term $I = \int_0^t \alpha(t, s) \frac{\delta \psi_t^{(n+1, m-1)}}{\delta z_s^*} ds$, we obtain

$$\begin{aligned} I &= (-1)^{m-1} \sum_{l=1}^{n+1} \int_{C_{n+1}} \frac{1}{(n+1)!} d^{n+1} \tau \int_{0 < s_1 < \dots < s_{2m-2} < t} \alpha(t, \tau_l) f_R(\tau, s, l) d^{2m-2} s, \\ f_R(\tau, s, l) &= \left(\prod_{k \neq l} z_{\tau_k}^* \right) \mathcal{L}_b(s_1, \dots, s_{2m-2}) \times \mathcal{T} \left(\prod_{k=1}^{n+1} L(\tau_k) \prod_{j=1}^{m-1} \left[L(s_{P(2j-1)}) L^\dagger(s_{P(2j)}) \right] \right) \end{aligned} \quad (23)$$

Consistency

So

RHS_2

$$\begin{aligned}
 &= -L^\dagger(t) \int_0^t \alpha(t, s) \frac{\delta \psi_t^{(n+1, m-1)}}{\delta z_s^*} ds \\
 &= (-1)^m \int_{C_n} \frac{1}{(n+1)!} d^n \boldsymbol{\tau} \sum_{l=1}^{n+1} \int_0^t d\tau_l \int_{0 < s_1 < \dots < s_{2m-2} < t} \alpha(t, \tau_l) f_R(\boldsymbol{\tau}, \mathbf{s}, l) d^{2m-2} \mathbf{s}, \\
 &= (-1)^m \int_{C_n} \frac{1}{n!} d^n \boldsymbol{\tau} \int_0^t d\tau' \int_{0 < s_1 < \dots < s_{2m-2} < t} f(\boldsymbol{\tau}, \mathbf{s}, \tau') d^{2m-2} \mathbf{s}, \\
 & f(\boldsymbol{\tau}, \mathbf{s}, \tau') = \left(\prod_{k=1}^n z_{\tau_k}^* \right) \mathcal{L}_b(s_1, \dots, s_{2m-2}) \alpha(t, \tau') \\
 & \times \mathcal{T} \left(\prod_{k=1}^n L(\tau_k) \prod_{j=1}^{m-1} [L(s_{P(2j-1)}) L^\dagger(s_{P(2j)})] L^\dagger(t) L(\tau') \right) \psi_0.
 \end{aligned}$$

(24)

Consistency

By inserting the integration over τ' into the time-ordered product, we have

$$\int_0^t d\tau' \int_{0 < s_1 < \dots < s_{2m-2} < t} = \sum_{l=1}^{2m-1} \int_{0 < s_1 < \dots < s_{l-1} < \tau' < s_l < \dots < s_{2m-2} < t}, \quad (25)$$

Collection all $2m - 1$ subdomains together, we obtain

$$\begin{aligned} RHS_2 &= (-1)^m \sum_{l=1}^{2m-1} \int_{C_n} \frac{1}{n!} d^n \tau \\ &\int_{0 < s_1 < \dots < s_{2m-1} < t} \alpha(t, s_l) \mathcal{L}_b(s_1, \dots, \hat{s}_l, \dots, s_{2m-1}) \\ &\times \mathcal{T} \left(L^\dagger(t) L(s_l) \prod_{k=1}^n L(\tau_k) \prod_{j=1}^{m-1} [L(s_{P(2j-1)}) L^\dagger(s_{P(2j)})] \right) \psi_0 d^{2m-1} s \\ &= B_{n,m}. \end{aligned} \quad (26)$$

Direct Discretization

NMQSD:

$$\frac{d}{dt}\psi_t = \left(-iH + Lz_t^* - L^\dagger \int_0^t \alpha(t, s) \frac{\delta}{\delta z_s^*} ds \right) \psi_t. \quad (27)$$

Discretization: Given

$$t_j = j\Delta t, z_j = z_{t_j}, \alpha_{i,j} = \alpha(t_i, t_j), \quad j = 0, 1, \dots, N, \quad (28)$$

we discrete the NMQSD equation as

$$\frac{\psi_{n+1} - \psi_n}{\Delta t} = (-iH + Lz_n^*) \psi_n - L^\dagger \sum_{j=0}^n \alpha_{n,j} \frac{\delta \psi_n}{\delta z_j^*}, \quad (29)$$

Example:

- $\psi_1 = (I - iH\Delta t + Lz_0^*\Delta t) \psi_0 = G_0\psi_0,$
- $\psi_2 = (I - iH\Delta t + Lz_0^*\Delta t) \psi_1 - L^\dagger \alpha_{1,0} \frac{\delta \psi_1}{\delta z_0^*} \Delta t$
 $= G_1 G_0 \psi_0 - (\Delta t)^2 \alpha_{1,0} L^\dagger L \psi_0 \psi_0.$

Direct Discretization

- $\psi_3 = G_2 G_1 G_0 \psi_0 - (\Delta t)^2 (\alpha_{1,0} G_2 L^\dagger L + \alpha_{2,0} L^\dagger G_1 L + \alpha_{2,1} L^\dagger L G_0) \psi_0,$
- $\psi_4 = G_3 G_2 G_1 G_0 \psi_0$
 $- (\Delta t)^2 (\alpha_{1,0} G_3 G_2 L^\dagger L + \alpha_{2,0} G_3 L^\dagger G_1 L + \alpha_{2,1} G_3 L^\dagger L G_0 -$
 $\alpha_{3,0} L^\dagger G_2 G_1 L - \alpha_{3,1} L^\dagger G_2 L G_0 - \alpha_{3,2} L^\dagger L G_1 G_0) \psi_0$
 $+ (\Delta t)^4 (\alpha_{2,1} \alpha_{3,0} L^\dagger L^\dagger L L + \alpha_{3,1} \alpha_{2,0} L^\dagger L^\dagger L L + \alpha_{3,2} \alpha_{1,0} L^\dagger L L^\dagger L) \psi_0$
- ...

Infeasible due to the explosion of terms!

Numerical Framework for Linear NMQSD

Core Methodology

- **Approach:** Iterative evolution of the wave function over discrete time steps.
- **Key Idea:** Explicitly resolve functional derivatives via **auxiliary states**, denoted by the superscript index.

Auxiliary Wave Functions: ψ_t^A

Let the superscript $A = \{s_1, \dots, s_k\}$ be the set of active insertion times.

- **Physical State:** $\psi_t^\emptyset \equiv \psi_t$ (Empty set $\emptyset$).
- **Auxiliary State:** Represents higher-order functional derivatives:

$$\psi_t^A \sim \frac{\delta^k}{\delta z_{s_1}^* \dots \delta z_{s_k}^*} \psi_t, \quad A = \{s_1, \dots, s_k\}.$$

Time Evolution ($t_n \rightarrow t_{n+1}$)

The state update accumulates contributions from three processes:

- 1. **Inheritance (Evolution)**

Preserves the memory index A .

$$\overline{\hspace{1.5cm}}^{G_n} \quad \psi_{n+1}^A = G_n \psi_n^A = (I - iH\Delta t + Lz_n^* \Delta t) \psi_n^A.$$

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- 2. **Insertion (Memory Creation)**

Index expands: $A \rightarrow A \cup \{t\}$.

$$\text{---} \overset{L\Delta t}{\text{---} \bigcirc \text{---}} \text{---} \quad \psi_{n+1}^{A \cup \{t_n\}} = \Delta t L \psi_n^A.$$

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- 3. Pairing (Memory Contraction)**

Index contracts: $A \rightarrow A \setminus \{s\}$.

$$\overset{-\alpha(t_n, s)L^\dagger \Delta t}{\text{---} \cdots \bullet \cdots \text{---}} \quad \psi_{n+1}^{A \setminus \{s\}} \leftarrow \psi_{n+1}^{A \setminus \{s\}} - \Delta t \alpha(t_n, s) L^\dagger \psi_n^A.$$

Examples

----- $t_0 \rightarrow t_1$ -----

$$\psi_1 = \text{thick bar} = \overset{G_0}{\text{thin bar}} = G_0 \psi_0$$

$$\psi_1^{\{0\}} = \text{thick bar with circle} = \overset{L\Delta t}{\text{thin bar with circle}} = L\psi_0\Delta t$$

Examples

----- $t_0 \rightarrow t_1$ -----

$$\psi_1 = \text{[thick bar]} = \text{[thin bar]}^{G_0} = G_0 \psi_0$$

$$\psi_1^{\{0\}} = \text{[thick bar with circle]} = \text{[thin bar with circle]}^{L\Delta t} = L\psi_0 \Delta t$$

----- $t_1 \rightarrow t_2$ -----

$$\psi_2 = \text{[thick bar]} = \text{[thick bar]}^{G_1} + \text{[thick bar with circle]}^{\overset{-\alpha_{1,0} L^\dagger \Delta t}{\text{[thick bar with filled circle]}}}$$

$$= G_1 \psi_1 - \alpha_{1,0} L^\dagger \Delta t \psi_1^{\{0\}}$$

$$\psi_2^{\{0\}} = \text{[thick bar with circle]} = \text{[thick bar with circle]}^{G_1} = G_1 \psi_1^{\{0\}}$$

$$\psi_2^{\{1\}} = \text{[thick bar]} = \text{[thick bar]}^{L\Delta t} = L\psi_1 \Delta t$$

$$\psi_2^{\{0,1\}} = \text{[thick bar with two circles]} = \text{[thick bar with circle]}^{L\Delta t} = L\psi_1^{\{0\}} \Delta t$$

Noise Generation

Colored Noise Generation

Input: a given bath correlation function $\alpha(t, s)$.

Output: the colored noise z_t with correlation $\mathbb{E}[z_t z_s^*] = \alpha(t, s)$.

1. Discretization $\Sigma_{i,j} = \alpha(t_i, t_j)$;
2. Cholesky decomposition: $\Sigma = L_c L_c^\dagger$;
3. Generate independent standard complex normal random variables $\xi_1, \xi_2, \dots, \xi_N \in \mathbb{C}$.
4. Compute the colored noise vector:

$$\mathbf{z} = L_c \boldsymbol{\xi}, \quad \text{where } \boldsymbol{\xi} = (\xi_1, \xi_2, \dots, \xi_N)^\top. \quad (30)$$

Then $\mathbb{E}(\mathbf{z}\mathbf{z}^*) = L_c \mathbb{E}(\boldsymbol{\xi}\boldsymbol{\xi}^*) L_c^\dagger = \Sigma$.

Computational Cost

Setup: Simulate M stochastic trajectories over N time steps.

The "Curse" of Memory

At the k -th step, the number of diagrams grows to 2^k .

- **Space Complexity:** $\mathcal{O}(2^N)$ (Exponential memory usage)
- **Time Complexity:** $\mathcal{O}(M \cdot N \cdot 2^N)$

Computational Cost

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Cost Reduction Strategies

- **Memory Cutoff (Finite Memory):**
Since $\alpha(t, s)$ decays, ignore arcs with length $> \tau_{mem}$.
- **Perturbation Truncation:**
In weak coupling regime (small L), drop diagrams with many insertion nodes ($\circ$).

Numerical Examples

Spin-boson model:

- System Hamiltonian: $H = \frac{\omega}{2}\sigma_z$, $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ is the Pauli Z matrix.
- Coupling operator: $L = \lambda\sigma_z$.
- Bath correlation function:

$$\alpha(\tau) = \frac{\gamma}{2}e^{-\gamma|\tau|}.$$

The exact solution is known:

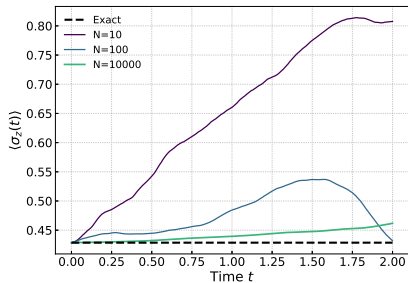
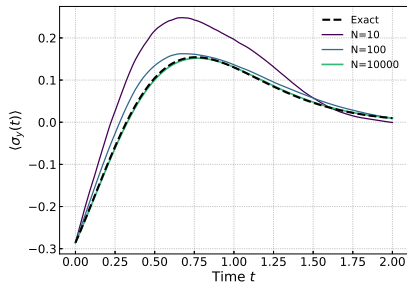
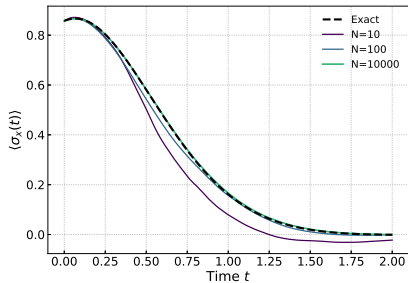
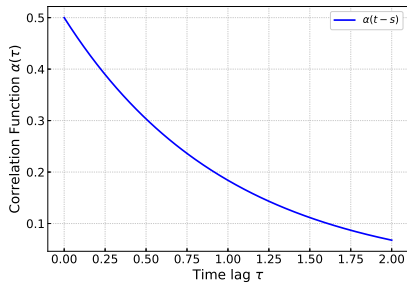
$$\rho(t) = \begin{pmatrix} \rho_{11}(0) & \rho_{12}(0)e^{-F(t)} \\ \rho_{21}(0)e^{-F(t)^*} & \rho_{22}(0) \end{pmatrix},$$

with

$$F(t) = i\omega t + \frac{2\lambda^2}{\gamma}(e^{-\gamma t} + \gamma t - 1).$$

Parameters: $\omega = 1$, $\lambda = \sqrt{2}$, $\gamma = 1$.

Ensemble averages



Numerical Examples

Spin-boson model:

- System Hamiltonian: $H = \epsilon\sigma_z + \Delta\sigma_x$, with $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.
- Coupling operator: $L = \sigma_x$.
- Bath correlation function:

$$\alpha(\tau) = \frac{1}{2} \sum_{j=1}^L \frac{c_j^2}{\omega_j} \left(\coth\left(\frac{\beta\omega_j}{2}\right) \cos(\omega_j\tau) - i \sin(\omega_j\tau) \right)$$

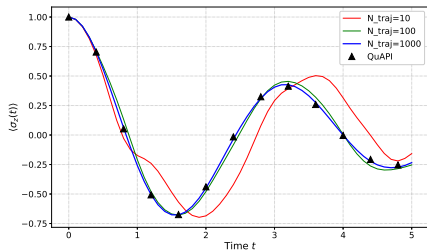
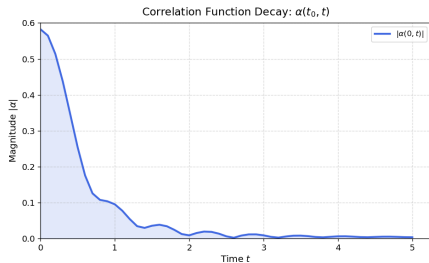
with Ohmic spectral density

$$\omega_j = -\omega_c \ln\left(1 - \frac{j}{L}(1 - \exp(-\omega_{\max}/\omega_c))\right),$$

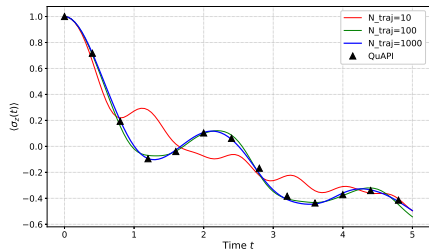
$$c_j = \omega_j \sqrt{\frac{\xi\omega_c}{L}(1 - \exp(-\omega_{\max}/\omega_c))}.$$

Parameters: $\beta = 5$, $\omega_c = 2.5\Delta$, $\xi = 0.2$, $L = 400$, $\omega_{\max} = 4\omega_c$.

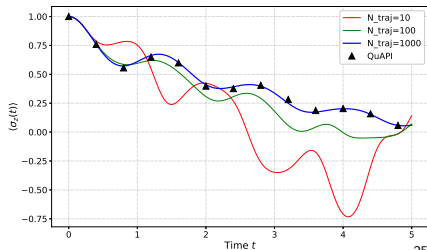
Ensemble averages



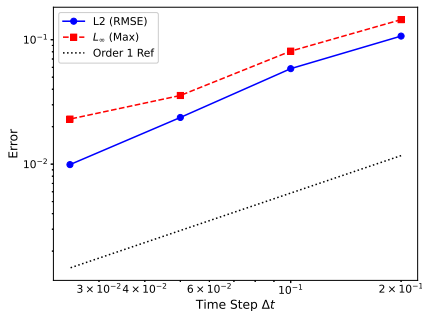
Bath Correlation $\alpha(\tau)$



$\varepsilon = 0$

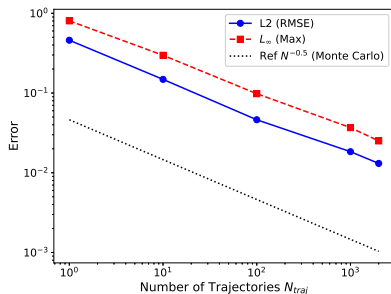


Convergence tests



Convergence w.r.t. time step Δt

(using $N_{\text{traj}} = 10^5$, $K\Delta t = 0.6$.)



Convergence w.r.t. number of trajectories

N_{traj}

(using $\Delta t = 0.05$.)

Conclusion

Summary:

- Series-form analytical solution of NMQSD.
- Iterative numerical method based on diagrammatic representation.

Future work:

- Higher-order schemes.
- Machine learning methods.

Thank you for your attention!